Healthcare Data Analysis & Dashboarding

Technical Implementation & Executive Insight Report

Dataset Baseline: 08-Jun-2023 **Total Reconciled Records:** 2,335 Patients **Primary Key:** Customer ID

Section 1: Executive Summary

Project Objective: To clean, transform, and analyze multi-source healthcare data across Medical Examinations, Hospitalization Details, and Customer Profiles. The primary focus is evaluating cost drivers, risk factors, and patient demographics using advanced Excel formulas, lookup functions, AI Quick Analysis tools, and dynamic dashboards.

Scope of Work Overview:

- **Data Cleaning:** Audited missing values (`?`), executed mode/mean imputations, and removed 8 invalid/orphan records.
- **Data Transformation:** Parsed customer names, standardized surgical metrics, calculated age relative to baseline (08-Jun-2023), assigned BMI diagnostic categories, and standardized financial metrics.
- **Data Integration:** Merged source tables into a unified master worksheet (`Healthcare`) via relational lookups.
- **Manual & AI Analysis:** Built cross-tabulation Pivot Tables, Donut and Column charts, AI Quick Analysis metrics, and an executive dashboard with interactive slicers.

TOTAL PATIENTS	TOTAL HEALTHCARE COST	AVG. HOSPITAL COST	AVERAGE AGE	DATA PURGE
2,335	$31.59M	$13,529.92	39.2	8 Rows
Cleaned Join	$31,592,358.61	Per Patient	Years Baseline	Invalid/Orphan

Section 2: Comprehensive Step-by-Step Implementation

Phase 1: Data Cleaning

Step 1.1: Missing Value Audit & Reconciliation (Removal of 8 Invalid Rows)

What: Audited all sheets for placeholder symbols (?) and un-linkable customer records.

Why: Reconciled row discrepancies between source tables to ensure a clean 1:1 join across all datasets.

How: Identified 8 invalid records in *Hospitalisation Details* (6 rows with Customer ID = "?", and 2 orphan rows with Customer ID = "`id2444`" and "`id3444`"). Deleted these 8 rows, reducing total rows from 2,343 to 2,335 to align perfectly with Customer Names and Medical Examinations.

Step 1.2: Imputing Date Values (Month & Year)

What: Filled missing month entries with "`Sep`" and missing year entries with the dataset's rounded average year.

Why: Prepares complete temporal records required for conversion into structured date objects.

Calculation & Formula: Mean Year = `1982.84` → **1983**.

```
=IF(B2="?", ROUND(AVERAGE($B$2:$B$2336), 0), B2)
```

Step 1.3: Mode Imputation for Categorical Fields

What: Identified statistical mode for `smoker`, `Hospital tier`, and `City tier`, replacing missing ? entries.

Why: Preserves categorical distributions without introducing bias or breaking group breakdowns.

Imputed Values: `smoker`: "No" (1,845 occurrences) | `Hospital tier`: "tier - 2" (1,339) | `City tier`: "tier - 2" (812).

```
=INDEX($G$2:$G$2336, MODE(MATCH($G$2:$G$2336, $G$2:$G$2336, 0)))
```

Step 1.4: State ID Standardization

What: Replaced missing State ID entries (?) with "Unknown".

Why: Prevents breaking lookup formulas, pivot grouping, and dynamic dashboard slicers.

```
=IF(I2="?", "Unknown", I2)
```

Phase 2: Data Transformation

Step 2.1: Splitting Customer Names

What: Separated full name strings (e.g., "Hawks, Ms. Kelly") into Title, First Name, and Last Name.

Why: Normalizes customer identity metrics for structured relational queries and clean table views.

Execution: Applied Excel Text-to-Columns feature delimited by comma and dot.

Step 2.2: Cleaning Major Surgeries Count

What: Converted `NumberOfMajorSurgeries` into numeric format by replacing "No major surgery" with `0`.

Why: Enables numerical aggregations (sum, average, risk-correlations) in pivot tables and models.

```
=IF(G2="No major surgery", 0, VALUE(G2))
```

Step 2.3: Consistency Check on Categorical Flags

What: Standardized text casing across `Heart Issues` and `smokers` columns.

Why: Eliminates duplicate grouping rows in pivot tables caused by casing variations (e.g., "yes", "YES", "Yes").

```
=PROPER(TRIM(Cell))
```

Step 2.4: Weight Status Classification

What: Created calculated field "Weight Status" based on Body Mass Index (BMI).

Why: Categorizes continuous numerical BMI metrics into standard clinical diagnostic categories.

```
=IF(BMI_Cell<18.5, "Underweight", IF(BMI_Cell<=24.9, "Normal Weight", IF(BMI_Cell<=29.9,
"Overweight", "Obesity")))
```

Step 2.5: Generating Date of Birth

What: Concatenated day, month, and year into a formatted Date of Birth field (`DD-MMM-YYYY`).

Why: Unifies individual date components into a standard serial date object for formula calculations.

```
=DATEVALUE(CONCATENATE(D2, "-", C2, "-", B2)) (Formatted via Custom Format: DD-MMM-YYYY)
```

Step 2.6: Calculating Patient Age

What: Calculated patient age relative to baseline date (8th June 2023).

Why: Enables demographic age grouping and regression against healthcare expenditure.

```
=DATEDIF(DOB_Cell, DATE(2023, 6, 8), "Y")
```

Step 2.7: Currency Formatting

What: Formatted healthcare expenditure charges as Currency (`$#,##0.00`).

Why: Ensures standardized financial presentation and alignment across analytical outputs.

Phase 3: Master Table Integration ("Healthcare")

Step 3.1: Unified "Healthcare" Master Table Creation

Created a central worksheet named `Healthcare`, merging all three source tables into a consolidated master table containing 2,335 rows using `Customer ID` as the primary key.

```
=VLOOKUP(A2, 'Customer Names'!$A:$D, 3, FALSE)
```

Field #	Master Column Name	Data Source	Data Type / Format
1	Customer ID	Primary Key	Text / Key String
2	First Name	Customer Names	Text String
3	BMI	Medical Examinations	Decimal (Numeric)
4	HBA1C	Medical Examinations	Decimal (Numeric)
5	Heart Issues	Medical Examinations	Categorical (Yes/No)
6	Any Transplants	Medical Examinations	Categorical (Yes/No)
7	Cancer History	Medical Examinations	Categorical (Yes/No)
8	Number of Major Surgeries	Medical Examinations	Integer (Numeric)
9	Smoker	Medical Examinations	Categorical (Yes/No)
10	Weight Status	Calculated Field	Diagnostic Category
11	Diabetes Status	Medical Examinations	Categorical (Yes/No)
12	Date of Birth	Calculated Field	Date (DD-MMM-YYYY)
13	Charges	Hospitalisation Details	Currency ($#,##0.00)
14	Hospital Tier	Hospitalisation Details	Categorical (Tier 1-3)

Field #	Master Column Name	Data Source	Data Type / Format
15	City Tier	Hospitalisation Details	Categorical (Tier 1-3)
16	State ID	Hospitalisation Details	Text Code
17	Age	Calculated Field	Integer (Years)

Phase 4: AI & Manual Visual Analysis and Dashboard Assembly

Step 4.1: Manual Pivot Table & Custom Visual Insights

1. Cancer History Distribution among Smokers vs. Non-Smokers (Donut Chart):

Objective: Evaluate relative cancer rates between active smokers and non-smokers.

Key Insight: Active smokers show a significantly higher proportion of recorded cancer cases compared to non-smokers, highlighting lifestyle smoking as a top-tier healthcare risk factor and cost driver.

2. Hospitalization Charges by Weight Status (Column Chart):

Objective: Compare total and average medical charges across BMI weight categories.

Key Insight: Patients in the Obesity category incur the highest total and average medical charges, driven by secondary health complications, extended hospital stays, and intensive care requirements.

Step 4.2: AI-Driven Summary Statistics & Quick Analysis

Execution: Highlighted master dataset and used Excel Quick Analysis Tools (`Ctrl + Q`) to generate summary totals, averages, and dynamic distributions.

Summary Metrics Table:

- **Total Customer Records:** 2,335
- **Total Hospitalization Records:** 2,335
- **Total Healthcare Expenditure:** $31,592,358.61
- **Average Cost Per Patient:** $13,529.92
- **Average Patient Age:** 39.2 Years

Step 4.3: Interactive Healthcare Dashboard Assembly

Layout Structure: Positioned KPI Summary Cards across the top header zone and arranged analytical charts in a balanced, aligned two-column grid below.

Interactive Controls: Integrated dynamic Excel Slicers connected across all pivot charts for *Weight Status* and *Diabetes Status*.

Formatting & Styling: Formatted header block as "Healthcare Dashboard - AI Analysis", updated chart color palettes for visual harmony, removed default gridlines, and enabled full cross-filtering via `PivotTable Analyze → Filter Connections`.